Lösungen · Quadratische Funktionen

1. $\boldsymbol{x = -2}$

2. $\mathbf{x_1 = -1;\ x_2 = -3}$ ..... $x^2 + 4x + 3 = 0;\ x = -2 \pm 1$

3. $\mathbf{N_1(x_1|0)\ mit\ x_1 = 3 - \sqrt{2} \approx 1{,}59;\ N_2(x_2|0)\ mit\ x_2 = 3 + \sqrt{2} \approx 4{,}41}$ ...
$x^2 - 6x + 7 = 0;\ x = 3 \pm \sqrt{2}$

4. $\mathbf{x_1 = 3;\ x_2 = -5}$ ..... $-x^2 - 2x + 5 = -10;\ x^2 + 2x - 15 = 0;\ x = -1 \pm 4$

5. **(1) wahr** ..... einzige Lösung $x = -7$
(2) falsch ..... $(x + 8)^2 = 16;\ x = -4$ oder $x = -12$
z. B. $\mathbf{x^2 + 1 = 0}$

6. $\mathbf{y = x^2 + 6x + 7}$ ..... $(x + 3)^2 - 2 = x^2 + 6x + 9 - 2$
$\mathbf{x_1 = -3 + \sqrt{2} \approx -1{,}59;\ x_2 = -3 - \sqrt{2} \approx -4{,}41}$ ...
$(x + 3)^2 = 2$